

Sean Luka Girgis

Senior PySpark / Python Data Engineer

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Professional Summary

Senior Data Engineer and software engineering professional with 20+ years of enterprise experience across Python, PySpark/Spark, Hadoop/Hive, SQL, Oracle, ETL pipelines, distributed processing, performance engineering, and backend integration patterns. Strong background building scalable data processing, reporting, telemetry, forecasting, and migration solutions using Python, SQL, PySpark, Pandas, Oracle, AWS data services, and production monitoring practices. Experienced documenting system designs, troubleshooting complex production issues, optimizing performance, collaborating across distributed teams, and applying AI/LLM tooling to data and automation workflows.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built automated Python/Pandas ETL pipelines ingesting P95 telemetry from 6,000+ infrastructure endpoints, replacing manual processes with repeatable, validated data workflows.
- Used PySpark/Spark patterns to process large-scale telemetry and support parallel forecasting workloads across high-volume infrastructure datasets.
- Built hybrid AWS/Oracle data platform patterns using S3 landing zones, AWS Glue ETL concepts, Redshift analytics workloads, and Oracle schemas for historical retention and reporting.
- Designed and optimized SQL/Oracle reporting schemas to support data warehousing, capacity planning, trend analysis, executive dashboards, and operational risk reporting.
- Integrated BMC TrueSight/TSCO, CA APM, and AppDynamics feeds into unified reporting datasets with reconciliation, validation, and data quality checks.
- Developed Prophet and scikit-learn forecasting models to predict capacity bottlenecks months ahead and support proactive remediation planning.
- Created dashboards and reporting outputs using Streamlit, matplotlib, seaborn, and plotly to communicate performance, utilization, and capacity insights to stakeholders.
- Consumed and integrated operational data from enterprise APIs and monitoring platforms, supporting automated reporting, troubleshooting, and data product delivery.
- Maintained documentation for data definitions, transformations, validation rules, operational workflows, and platform decisions to support governance and knowledge transfer.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, strengthening production visibility and performance analysis.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and provide actionable recommendations to engineering teams.
- Implemented End-User Monitoring and transaction tracing across web and mobile systems to improve visibility into user journeys, API behavior, and application performance.

- Integrated HP Performance Center with Dynatrace to correlate load-test behavior, APM telemetry, and infrastructure metrics.
- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, operational risks, and data requirements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large financial-services enterprise with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules used to monitor enterprise application performance and support operational decision-making.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and replace manual weekly exports.
- Partnered with J2EE/WebLogic and infrastructure teams to diagnose bottlenecks using APM telemetry, SQL/Oracle data, thread behavior, and infrastructure KPIs.
- Delivered technical documentation, onboarding guidance, and client training for monitoring standards, deployment processes, and production troubleshooting.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits, response-time degradation, and resource bottlenecks including JDBC connections, threads, heap memory, CPU, and garbage collection.
- Used Java/J2EE, JMX monitoring, Wily Introscope, and thread-dump automation practices to provide real-time visibility during performance validation.
- Documented performance baselines, regression thresholds, findings, and recommendations used by engineering teams for release-gate decisions.
- Collaborated with application, testing, and infrastructure teams to troubleshoot performance defects and validate remediation outcomes.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness across large transaction datasets during phased cutover.
- Automated schema conversion and data validation workflows to protect data integrity across relational database migration paths.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability telecom billing interfaces using C, C++, Pro*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and Unix environments.
- Improved throughput and database performance in billing processes through SQL tuning, sequence optimization, memory reduction, and interface-level performance analysis.
- Built and maintained reporting, invoicing, settlement, and integration components across full billing lifecycle systems for large-scale telecom operations.

- Worked with enterprise integration and messaging patterns including MQ Series, XML-based workflows, WebLogic/WebSphere, and Unix automation.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt / Cairo, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, C++, C, Java, Perl, Korn Shell, XML, JSON, JavaScript concepts

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a serverless data platform pattern using S3, Glue, Athena, Redshift, Bedrock, PySpark, Parquet, and Snappy compression for cloud-scale ingestion, transformation, query, and AI-enabled access.

Technologies: AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet, Bedrock

AI-Powered Job Search Pipeline

Built an agentic Python pipeline for job intake, semantic duplicate detection, LLM scoring, JSON artifact generation, quality gates, and tracking using FAISS, sentence-transformers, Grok API, and Claude Code workflows.

Technologies: Python, FAISS, Sentence Transformers, LLM APIs, RAG, Agentic Workflows, JSON Contracts

HorizonScale — AI Capacity Forecasting Engine

Designed an AI-driven forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform enterprise telemetry into capacity-risk predictions and operational decision support.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Telemetry Data, Forecasting

High-Volume Database Migration Validation

Built validation and performance-sensitive migration tooling for a high-throughput travel shopping engine moving from MySQL nodes to Oracle RAC.

Technologies: C++, OCCl, OCI, Oracle RAC, SQL, Performance Validation